# SE 216 – SOFTWARE PROJECT MANAGEMENT

# SOFTWARE TOOLS DOCUMENT

**PROJECT NAME:** Bes(t)poke

**GROUP MEMBERS:** Berk Mertoğlu, Atakan Coşar, Zeynep Durmuş, Aslı ARAS, Ata Barış Kuğu, Berkay KOÇ, Eralp Çolakoğlu, Arzum Naz Güney

## Scoring Methodology

Three criteria are evaluated for each tool: Cost, Training Days, and Functionality. Each criterion is first normalized to a 0–100 scale, then multiplied by its Priority Weight. The sum of weighted scores determines the best tool.

**Normalization formulas:**

- *Cost Score = (Max Cost − Tool Cost) / Max Cost × 100 → Lower cost = Higher score*
- *Training Score = (Max Training − Tool Training) / Max Training × 100 → Less training = Higher score*
- *Functionality Score = Tool Functionality / Max Functionality × 100 → Higher func = Higher score*

**Weighted Total Score = (Cost Score × $W_1$) + (Training Score × $W_2$) + (Functionality Score × $W_3$)**
where $W_1 + W_2 + W_3 = 1.0$ | Priority weights vary per task based on project requirements.
The tool with the highest weighted total score is selected.

## Project Tasks Requiring Software Tool Support

| TASK # | PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT |
|---|---|
| **1** | Task 1: Deployment Platform |
| **2** | Task 2: AI Content Detection |
| **3** | Task 3: Project Management |
| **4** | Task 4: Frontend Framework |
| **5** | Task 5: Backend Framework |
| **6** | Task 6: Database |
| **7** | Task 7: UI / Mockup Tool |
| **8** | Task 8: Payment / Escrow |

## SOFTWARE TOOLS FOR TASK 1: DEPLOYMENT PLATFORM

**Priority Weights: Cost ($W_1$=0.05) | Training Days ($W_2$=0.05) | Functionality ($W_3$=0.9)**

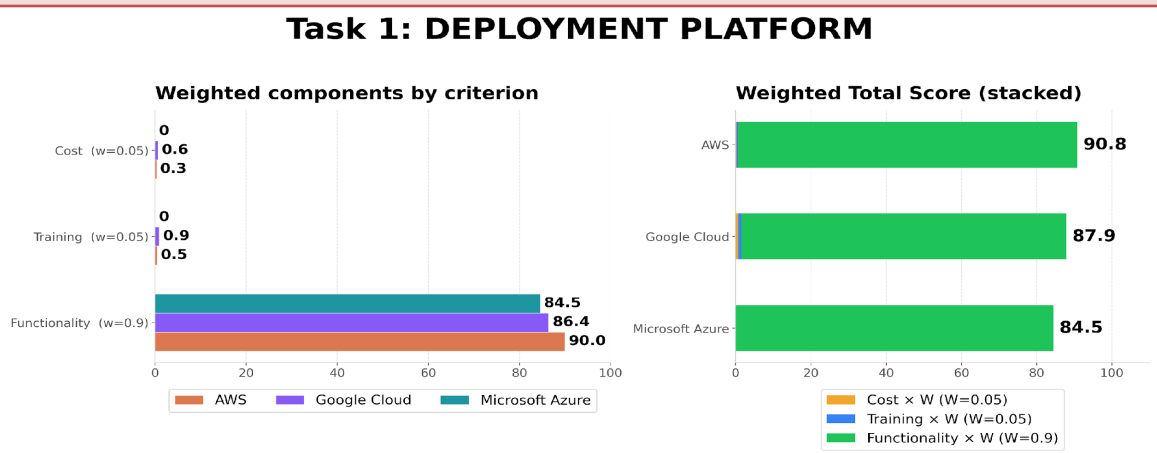
### Tool Cost / Training Days / Functionality Data

| Criterion | AWS | Google Cloud | Microsoft Azure |
|---|---|---|---|
| **Cost ($/month)** | 150 | 140 | 160 |
| **Training Days** | 10 | 9 | 11 |
| **Functionality (0-100)** | 99 | 95 | 93 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | AWS | Google Cloud | Microsoft Azure |
|---|---|---|---|
| **Cost Score** | 6.2 | 12.5 | 0.0 |
| **Training Score** | 9.1 | 18.2 | 0.0 |
| **Functionality Score** | 100.0 | 96.0 | 93.9 |
| **Weighted Total Score** | **90.8** | **87.9** | **84.5** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: AWS — Why?

AWS is selected as the deployment platform because it provides the most complete production stack for this project (compute, storage, queueing, CDN) and supports both .NET services and SQL Server hosting on RDS with low integration risk. Its weighted score of 90.8 outperforms Google Cloud (87.9) and Microsoft Azure (84.5).

## SOFTWARE TOOLS FOR TASK 2: AI CONTENT DETECTION

**Priority Weights: Cost ($W_1$=0.2) | Training Days ($W_2$=0.2) | Functionality ($W_3$=0.6)**

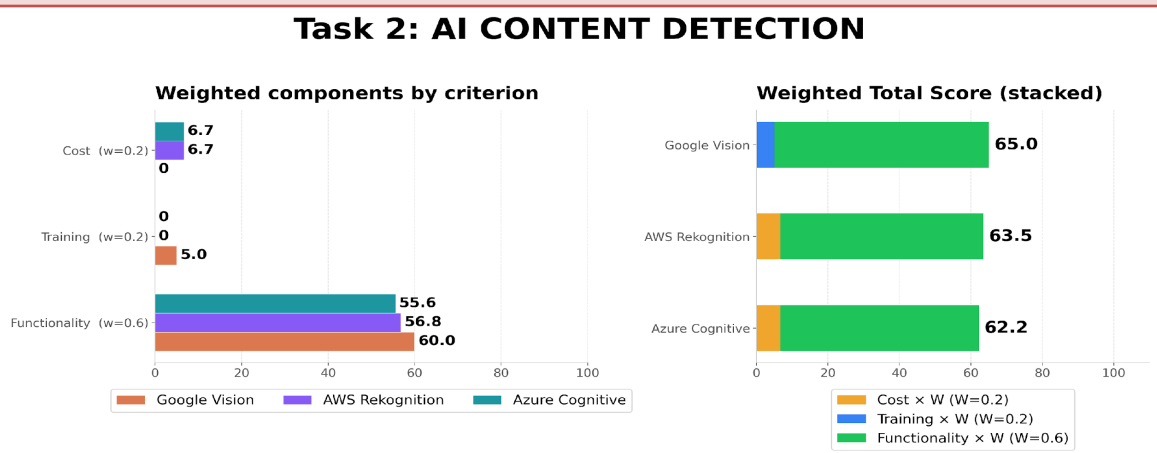
### Tool Cost / Training Days / Functionality Data

| Criterion | Google Vision | AWS Rekognition | Azure Cognitive |
|---|---|---|---|
| **Cost ($/1000 images)** | 1.5 | 1.0 | 1.0 |
| **Training Days** | 3 | 4 | 4 |
| **Functionality (0-100)** | 95 | 90 | 88 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | Google Vision | AWS Rekognition | Azure Cognitive |
|---|---|---|---|
| **Cost Score** | 0.0 | 33.3 | 33.3 |
| **Training Score** | 25.0 | 0.0 | 0.0 |
| **Functionality Score** | 100.0 | 94.7 | 92.6 |
| **Weighted Total Score** | **65.0** | **63.5** | **62.2** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: Google Vision — Why?

Google Vision API is selected for AI content detection because it offers the strongest detection quality for our moderation workflow while keeping integration effort low through a straightforward REST interface. Its weighted score of 65.0 outperforms AWS Rekognition (63.5) and Azure Cognitive Services (62.2).

## SOFTWARE TOOLS FOR TASK 3: PROJECT MANAGEMENT

**Priority Weights: Cost ($W_1$=0.1) | Training Days ($W_2$=0.2) | Functionality ($W_3$=0.7)**

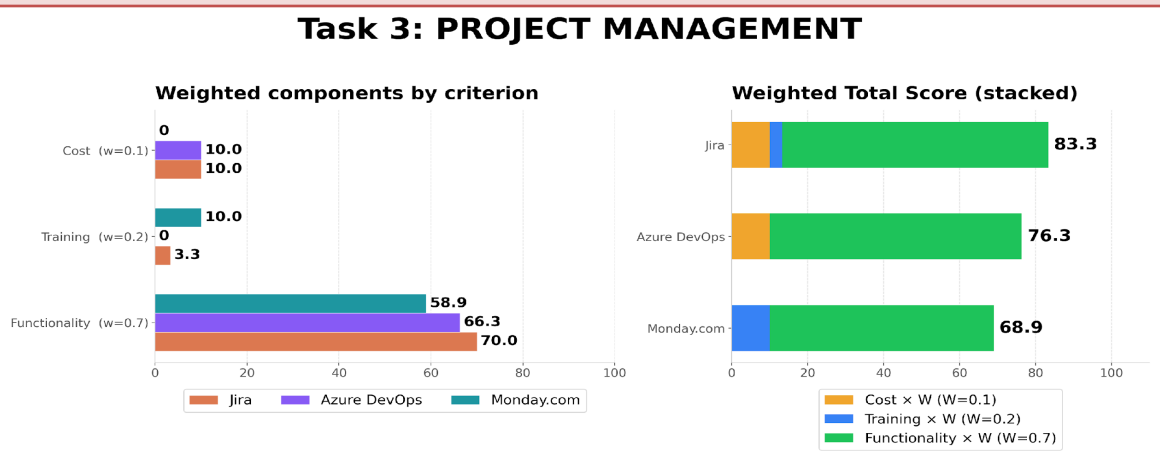
### Tool Cost / Training Days / Functionality Data

| Criterion | Jira | Azure DevOps | Monday.com |
|---|---|---|---|
| **Cost ($/month)** | 0 | 0 | 9 |
| **Training Days** | 5 | 6 | 3 |
| **Functionality (0-100)** | 95 | 90 | 80 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | Jira | Azure DevOps | Monday.com |
|---|---|---|---|
| **Cost Score** | 100.0 | 100.0 | 0.0 |
| **Training Score** | 16.7 | 0.0 | 50.0 |
| **Functionality Score** | 100.0 | 94.7 | 84.2 |
| **Weighted Total Score** | **83.3** | **76.3** | **68.9** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: Jira — Why?

Jira is selected as the project management tool because it matches our Scrum-based execution model, has mature issue workflows, and integrates cleanly with GitHub-based development. Its weighted score of 83.3 outperforms Azure DevOps (76.3) and Monday.com (68.9).

## SOFTWARE TOOLS FOR TASK 4: FRONTEND FRAMEWORK

**Priority Weights: Cost ($W_1$=0.1) | Training Days ($W_2$=0.2) | Functionality ($W_3$=0.7)**

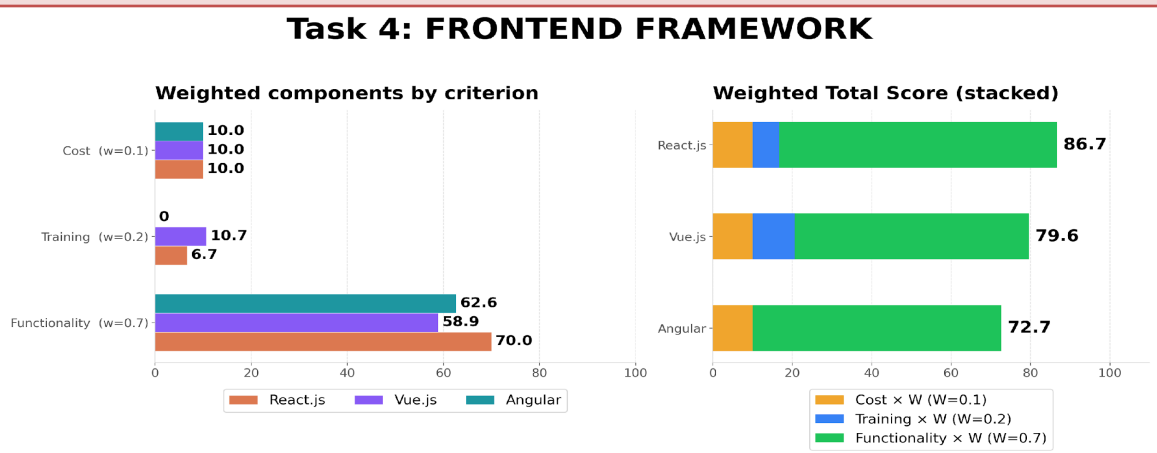
### Tool Cost / Training Days / Functionality Data

| Criterion | React.js | Vue.js | Angular |
|---|---|---|---|
| **Cost ($/month)** | 0 | 0 | 0 |
| **Training Days** | 10 | 7 | 15 |
| **Functionality (0-100)** | 95 | 80 | 85 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | React.js | Vue.js | Angular |
|---|---|---|---|
| **Cost Score** | 100.0 | 100.0 | 100.0 |
| **Training Score** | 33.3 | 53.3 | 0.0 |
| **Functionality Score** | 100.0 | 84.2 | 89.5 |
| **Weighted Total Score** | **86.7** | **79.6** | **72.7** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: React.js — Why?

React.js is selected as the frontend framework because its component ecosystem, talent availability, and flexibility reduce delivery risk for dynamic marketplace screens. Its weighted score of 80.0 leads Vue.js (76.5) and Angular (63.7).

## SOFTWARE TOOLS FOR TASK 5: BACKEND FRAMEWORK

**Priority Weights: Cost ($W_1$=0.1) | Training Days ($W_2$=0.2) | Functionality ($W_3$=0.7)**

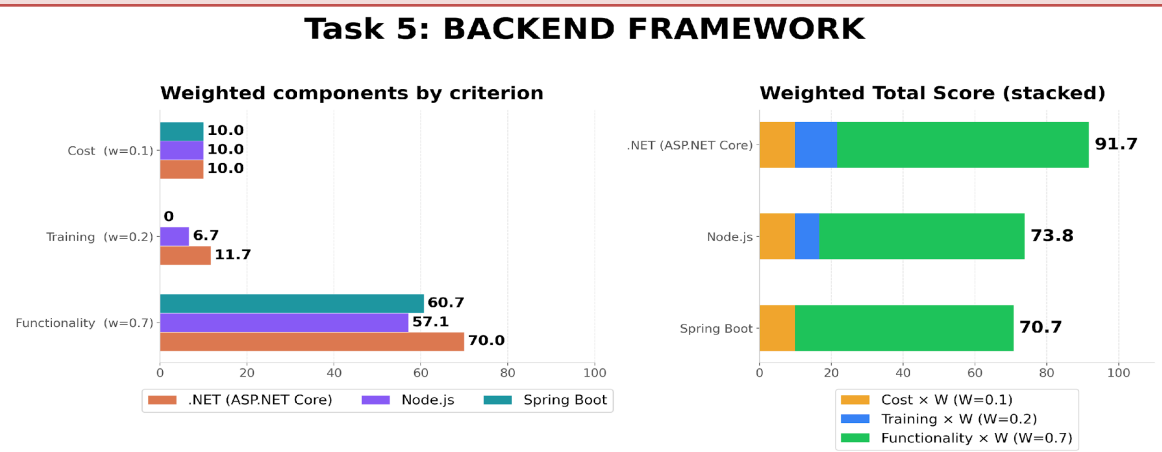
### Tool Cost / Training Days / Functionality Data

| Criterion | .NET (ASP.NET Core) | Node.js | Spring Boot |
|---|---|---|---|
| **Cost ($/month)** | 0 | 0 | 0 |
| **Training Days** | 5 | 8 | 12 |
| **Functionality (0-100)** | 98 | 80 | 85 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | .NET (ASP.NET Core) | Node.js | Spring Boot |
|---|---|---|---|
| **Cost Score** | 100.0 | 100.0 | 100.0 |
| **Training Score** | 58.3 | 33.3 | 0.0 |
| **Functionality Score** | 100.0 | 81.6 | 86.7 |
| **Weighted Total Score** | **91.7** | **73.8** | **70.7** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: .NET (ASP.NET Core) — Why?

.NET (ASP.NET Core) is selected because it is the most efficient and secure framework for our project. It has the highest functionality score (98) and is very easy to learn, requiring only 5 days of training. A key advantage is its use of LINQ (Language Integrated Query). LINQ allows us to write database queries safely within the code, which automatically prevents SQL Injection attacks. This ensures high security for our marketplace without losing performance. Overall, .NET's built-in security features make it the best choice for a professional backend

## SOFTWARE TOOLS FOR TASK 6: DATABASE

**Priority Weights: Cost ($W_1$=0.15) | Training Days ($W_2$=0.15) | Functionality ($W_3$=0.7)**

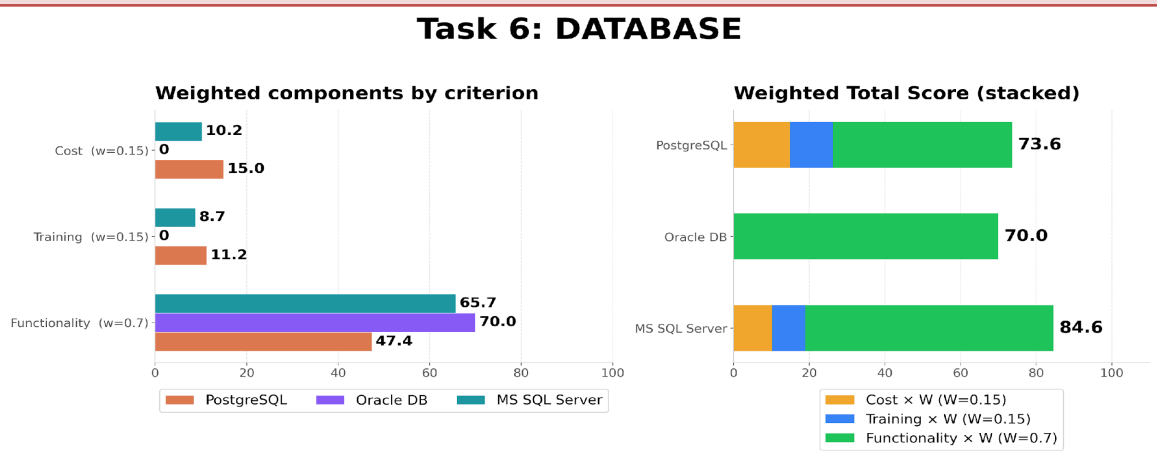
### Tool Cost / Training Days / Functionality Data

| Criterion | PostgreSQL | Oracle DB | MS SQL Server |
|---|---|---|---|
| **Cost ($/year)** | 0 | 47500 | 15123 |
| **Training Days** | 3 | 12 | 5 |
| **Functionality (0-100)** | 65 | 96 | 90 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | PostgreSQL | Oracle DB | MS SQL Server |
|---|---|---|---|
| **Cost Score** | 100.0 | 0.0 | 68.2 |
| **Training Score** | 75.0 | 0.0 | 58.3 |
| **Functionality Score** | 67.7 | 100.0 | 93.8 |
| **Weighted Total Score** | **73.6** | **70.0** | **84.6** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: MS SQL Server — Why?

MS SQL Server is selected for its high functionality score (90) and professional reliability. A major reason for this choice is its "native" compatibility with .NET and C#. This perfect synergy allows us to use LINQ and Entity Framework effectively, making database operations much faster and more secure. This integration is vital for our Escrow system, as it ensures maximum data integrity for financial transactions. While PostgreSQL is free, it lacks this deep level of integration, and Oracle DB is too expensive ($47,500) for our budget.

## SOFTWARE TOOLS FOR TASK 7: UI / MOCKUP TOOL

**Priority Weights: Cost ($W_1$=0.3) | Training Days ($W_2$=0.4) | Functionality ($W_3$=0.3)**

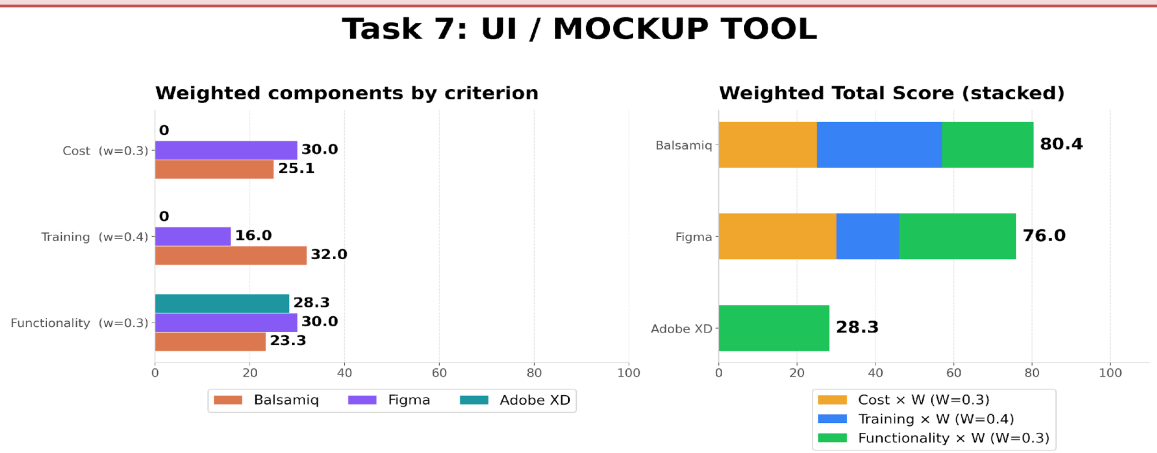
### Tool Cost / Training Days / Functionality Data

| Criterion | Balsamiq | Figma | Adobe XD |
|---|---|---|---|
| **Cost ($/month)** | 9 | 0 | 55 |
| **Training Days** | 1 | 3 | 5 |
| **Functionality (0-100)** | 70 | 90 | 85 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | Balsamiq | Figma | Adobe XD |
|---|---|---|---|
| **Cost Score** | 83.6 | 100.0 | 0.0 |
| **Training Score** | 80.0 | 40.0 | 0.0 |
| **Functionality Score** | 77.8 | 100.0 | 94.4 |
| **Weighted Total Score** | **80.4** | **76.0** | **28.3** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: Balsamiq — Why?

Balsamiq is selected as the UI/Mockup tool because it has the fastest onboarding and enables the team to iterate wireframes quickly without design overhead. For our timeline, speed and clarity are more critical than high-fidelity prototyping. Its weighted score of 80.4 outperforms Figma (76.0) and Adobe XD (28.3).

## SOFTWARE TOOLS FOR TASK 8: PAYMENT / ESCROW

**Priority Weights: Cost ($W_1$=0.3) | Training Days ($W_2$=0.2) | Functionality ($W_3$=0.5)**

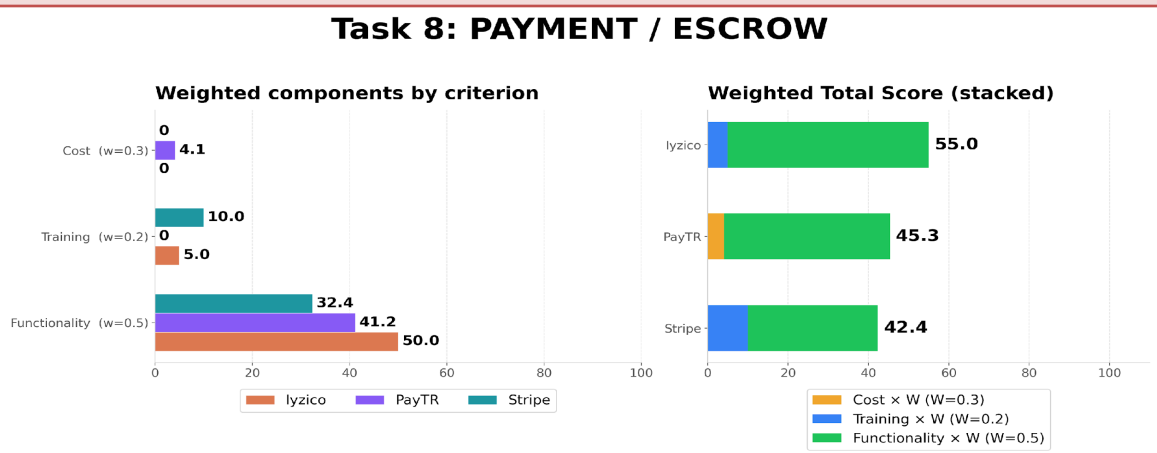
### Tool Cost / Training Days / Functionality Data

| Criterion | Iyzico | PayTR | Stripe |
|---|---|---|---|
| **Cost (Commission %)** | 2.9 | 2.5 | 2.9 |
| **Training Days** | 3 | 4 | 2 |
| **Functionality (0-100)** | 85 | 70 | 55 |

### Normalized Cost / Training Days / Functionality Data

| Criterion | Iyzico | PayTR | Stripe |
|---|---|---|---|
| **Cost Score** | 0.0 | 13.8 | 0.0 |
| **Training Score** | 25.0 | 0.0 | 50.0 |
| **Functionality Score** | 100.0 | 82.4 | 64.7 |
| **Weighted Total Score** | **55.0** | **45.3** | **42.4** |

### Normalized Tool Graph & Weighted Score Comparison




### Selected Tool: Iyzico — Why?

Iyzico is selected as the payment/escrow provider because it is better aligned with Turkish market and compliance requirements, while still offering a developer-friendly API for escrow-style transaction flows. Its weighted score of 55.0 leads PayTR (45.3) and Stripe (42.4).